Protecting the Fleet: Ground-Truth Economics Under NYC's Congestion Relief Charge

A Before-and-After Analysis to Preserve Manhattan Zone Coverage

Sing, Jennifer P. | sing.jenniferp@gmail.com

CONTEXT & LANDSCAPE

Two Miles or Twenty: One Flat Charge

A flat fee takes a bigger bite out of a short fare

Image: RPA.org | CRZ Fee Source: MTA

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Razor-Thin Operator Margins

Only

$4.60

per $100

Uber's adjusted EBITDA as a share of gross bookings, Q4 2025 (company-wide)

Source: Uber Technologies (2026)

Unstable Driver Supply

60%+

drivers quit

US Ride share drivers quit within their first 6 months

Source: Cook et al (2018)

Thin margins + drivers who can leave = a supply problem first, a pricing problem later.

EBITDA is Earnings Before Interest, Taxes, Depreciation, and Amortization

Body

Drivers Say It's Not Worth It. Is It?

Two ways to get this wrong: ignore a real problem, or pay for an imagined one

Short trips in the charged zone are no longer worth taking.

IS IT TRUE?

YES

1. Driver take-home pay falls
2. Short trips look less attractive
3. Coverage erodes
4. Waits rise, matches fall

NO: FALSE ALARM

No real problem: extra incentives would be money wasted

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Decision: intervene or not? In which zones? By how much?

Replacing Anecdotes with Empirical Proof

SMART deliverables mapped directly to the failure cascade below 60th Street

Driver Earnings Compression

Driver take-home pay falls

SMART Goal: Detect if income statistically dropped and identify economically marginal zones where net pay dropped post-charge (2024 vs. 2025) to target retention subsidies.

Short-Trip Resilience

Short trips look less attractive

SMART Goal: Detect statistically significant decrease in <2 mi ride share in the CRZ (2024 vs. 2025)

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Zone Coverage Resilience

Coverage Erodes

SMART Goal: Statistically evaluate relative trip volume decline in CRZ vs.NonCRZ (2024 vs. 2025)

Service Quality (SLA)

Waits rise, matches fall

SMART Goal: Detect statistically significant increases in wait times and declines in shared match rates inside the CRZ (2024 vs. 2025)

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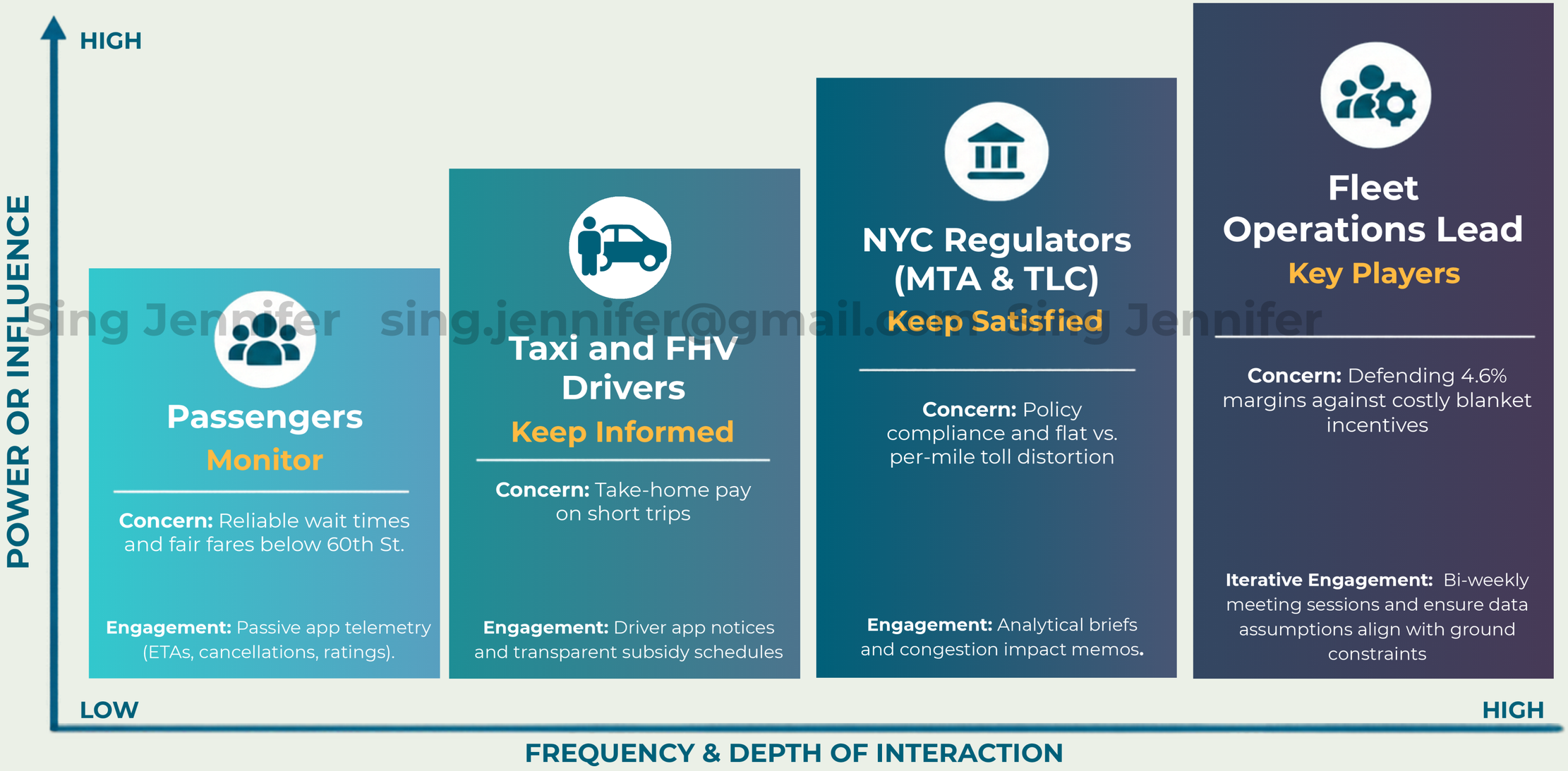
Aligning the Right Stakeholder with the Right Data

Tailoring engagement depth from passive app telemetry to bi-weekly executive co-design.



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Aligning Technical Goals with Business Impact

Evaluating Change Through Evidence-Based Frameworks

#	Technical objective	Evaluation criteria
T1	Integrate four TLC datasets	Unified table across all four services
T2	Validate data quality	Quality validation report
T3	Analyze pre- vs. post-charge trends	Documented and evaluated quality checks
T4	Test statistical significance	Hypothesis testing and ANOVA
T5	Estimate charge impact on driver earnings	Regression controlling for trip factors and seasonality
T6	Segment zones by economic impact	Actionable K-means clusters
T7	Communicate actionable insights	Evidence-based findings and recommendations

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617 Million Rides: Navigating the NYC TLC Dataset

Jan 2024 - Dec 2025
2- Year Data

Taxi Lookup Zone
38 charged | 225 control

NYC Taxi & Limousine Commission (TLC)
https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

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pickup/dropoff time and zone	✓	✓	✓	✓ Pickup zone ~80% null
trip_miles passenger_paid cbd_congestion_fee NULL for 2024 data	✓	✓	✓	
driver_pay request_ts shared_match			✓	

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Building the Baseline: Architecture, Logic, and Security

Transforming fragmented trip data into a governed, analytics-ready schema

Raw Data
Ingestion

- Original raw data from NYC Trip

Core
Standardisation

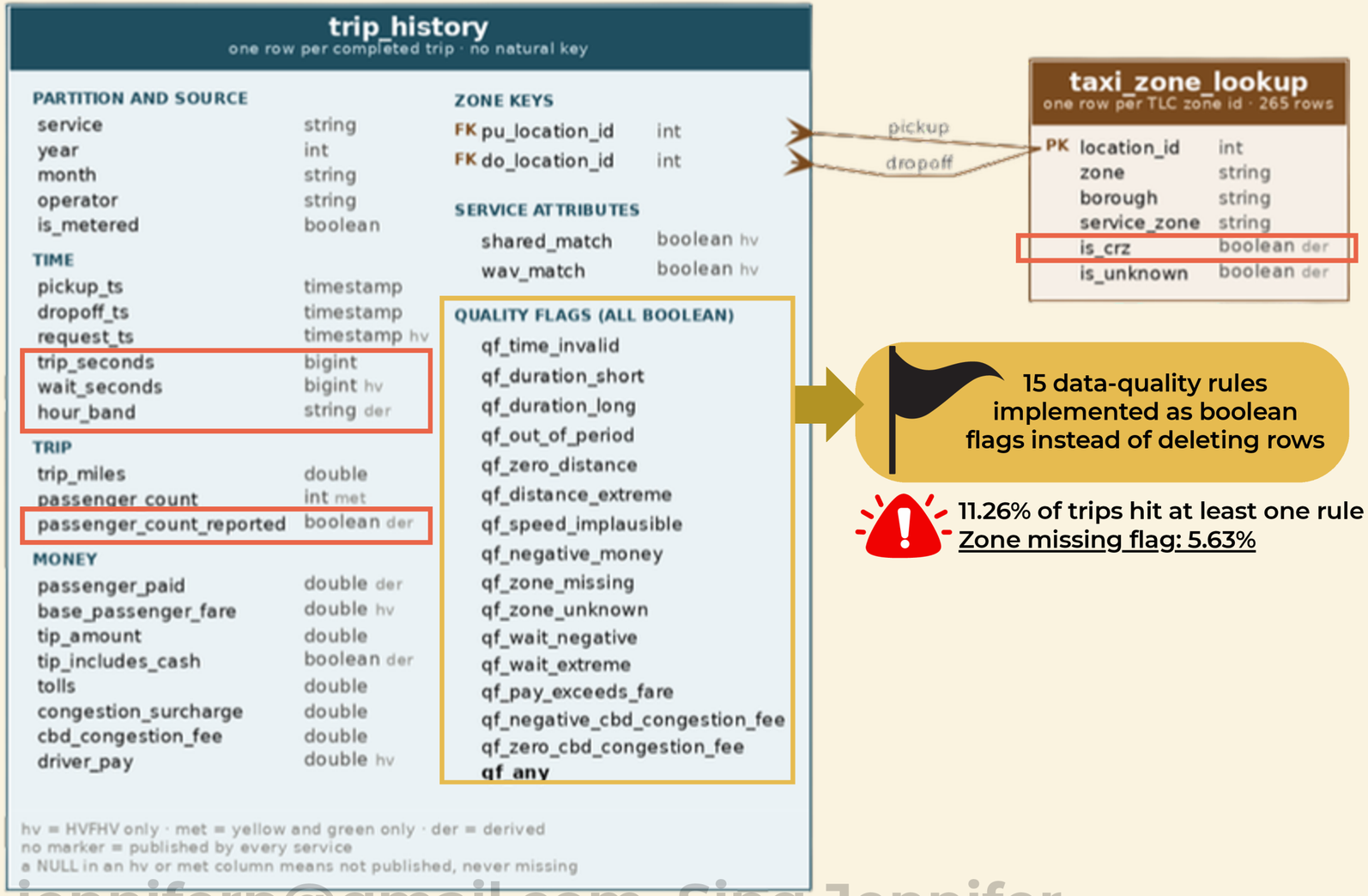
- Rename fields
- Cast data types
- Derive new fields

Conformed
Unification

- Unify equivalent fields.
- Create one analysis table.
- Add the zone reference.

Aggregate
Analytics Ready

- Aggregate fields for analytics purposes



15 data-quality rules implemented as boolean flags instead of deleting rows

11.26% of trips hit at least one rule
Zone missing flag: 5.63%

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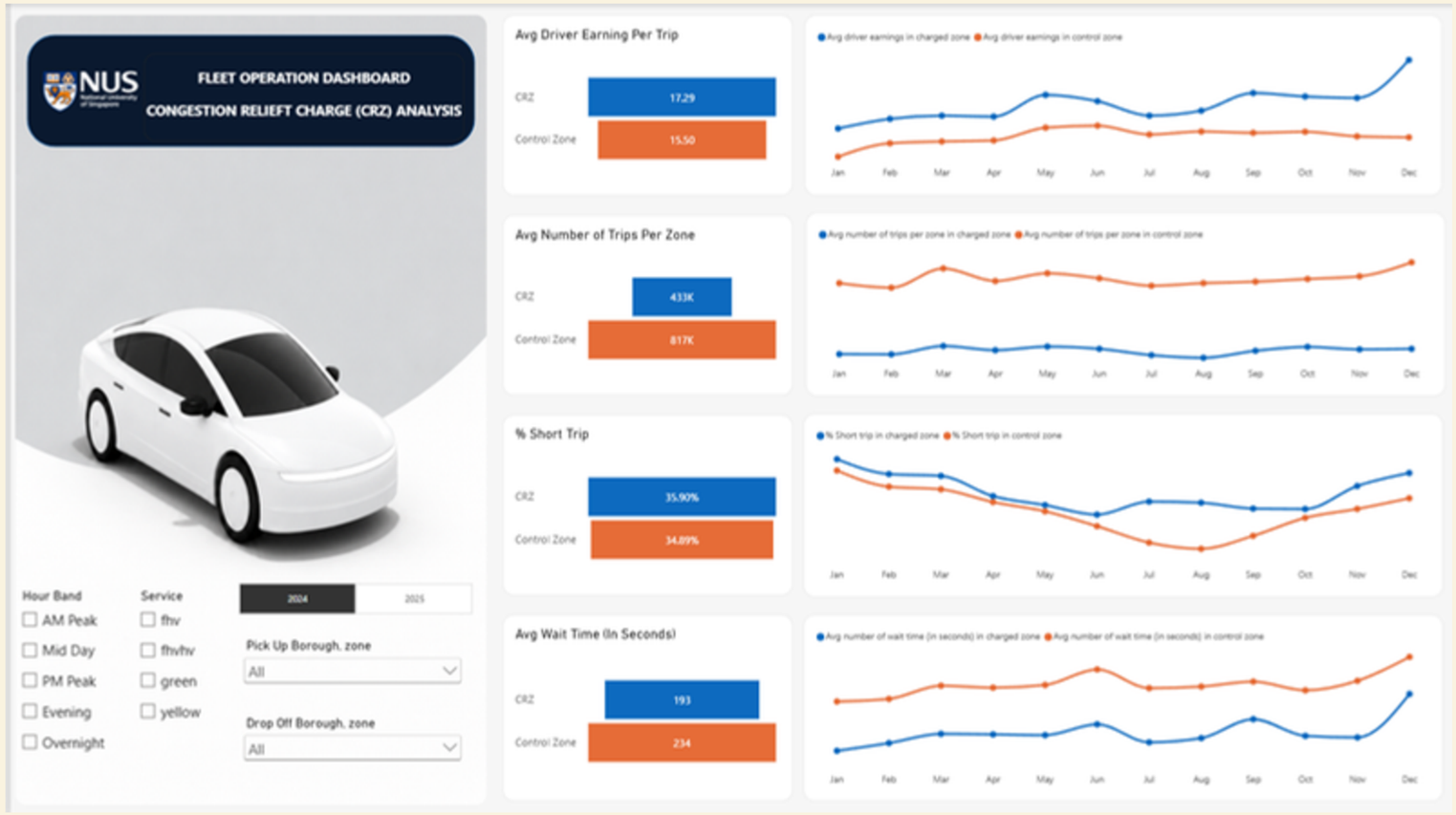
Visualising the behaviours caused by CRZ fee

A self-service reporting tool for business users of fleet operators



Driver earnings
Seemingly no adverse effect

Short Trip Volume
Seemingly no adverse effect

Wait Times
Rising in both charged and control zones.

Shared Rides
Seemingly no adverse effect

Historical trends are insufficient to isolate the impact of CRZ and make conclusion.

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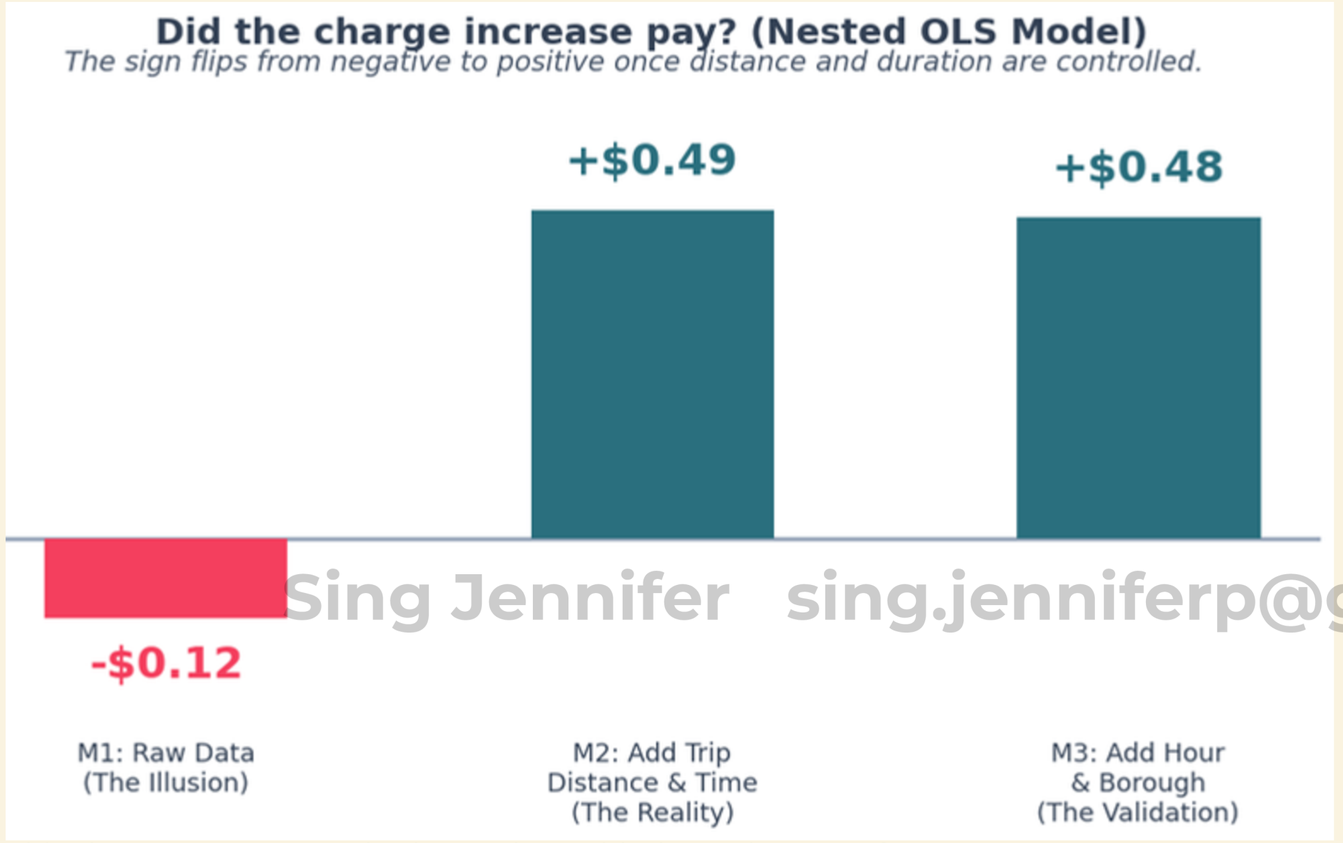
Statistical analysis is necessary to properly examine these relationships

The "Hidden" Raise: Per-Trip Revenue Impact (Regression Analysis)

Factoring in trip distance and duration reveals a true $0.48 per-trip pay increase



This chart was generated using Python with the assistance of GenAI

Statistical Illusion

Raw averages show a $0.12 drop; controls (time & distance) reveal a $0.49 increase per trip.

All Trip Lengths Win

Both short trips (<2 miles) and long trips (≥2 miles) saw positive pay growth (+$0.28 and + $0.48 respectively).

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Robust Validation

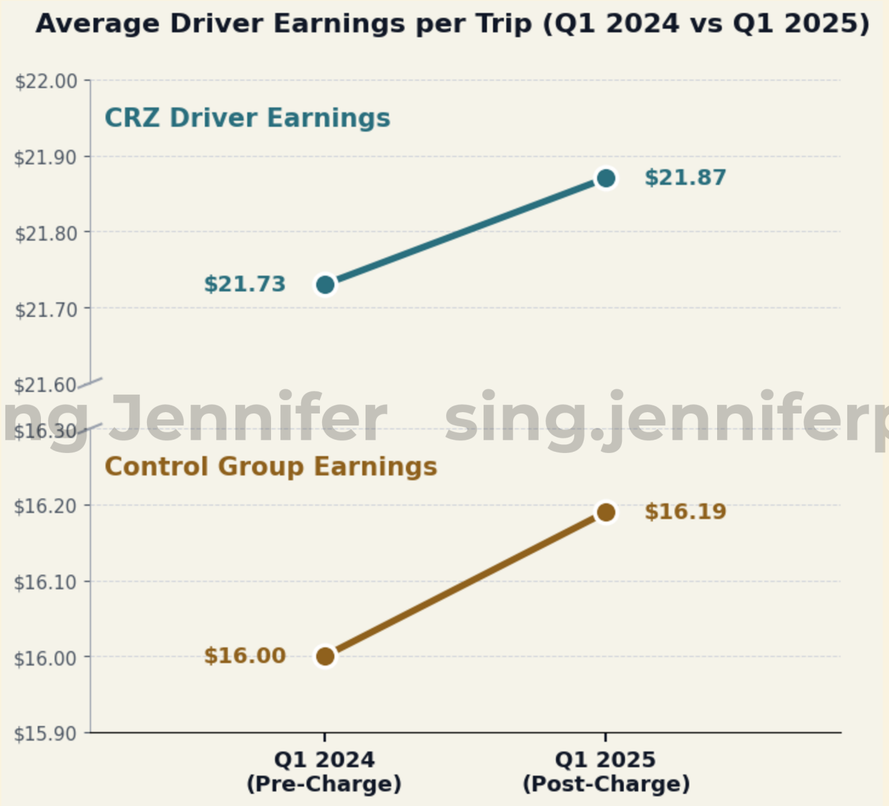
Time-of-day and borough controls confirm a stable +$0.48 pay increase per trip citywide.

Caveat: *Model isolates per-trip revenue, not total hourly earnings. Statistically validated for collinearity (VIF < 4) using robust standard errors*

Economic Impact: Driver Earnings Assessment

CBD Congestion Fee Shows No Negative Impact on Drivers Earnings

Driver Earnings



Q1 2024 vs Q1 2025

Excluded Jan 1-4

Test Group: CRZ Trips
Control Group: Non-CRZ Trips

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Scope note: A trip is considered inside the Manhattan Congestion Relief Zone (CRZ) if either the pickup (PU) location OR the drop-off (DO) location falls within the charged Manhattan zone boundaries.

Verdict: Stable. No Significant Decrease

Drivers are successfully absorbing the fee without trip-level pay cuts. Maintain current deployment patterns; wage-protection subsidies are not needed.

Test: Z-stat: 1.92 p: 0.9726 *Control Z-stat: 5.49 p: 1.00*

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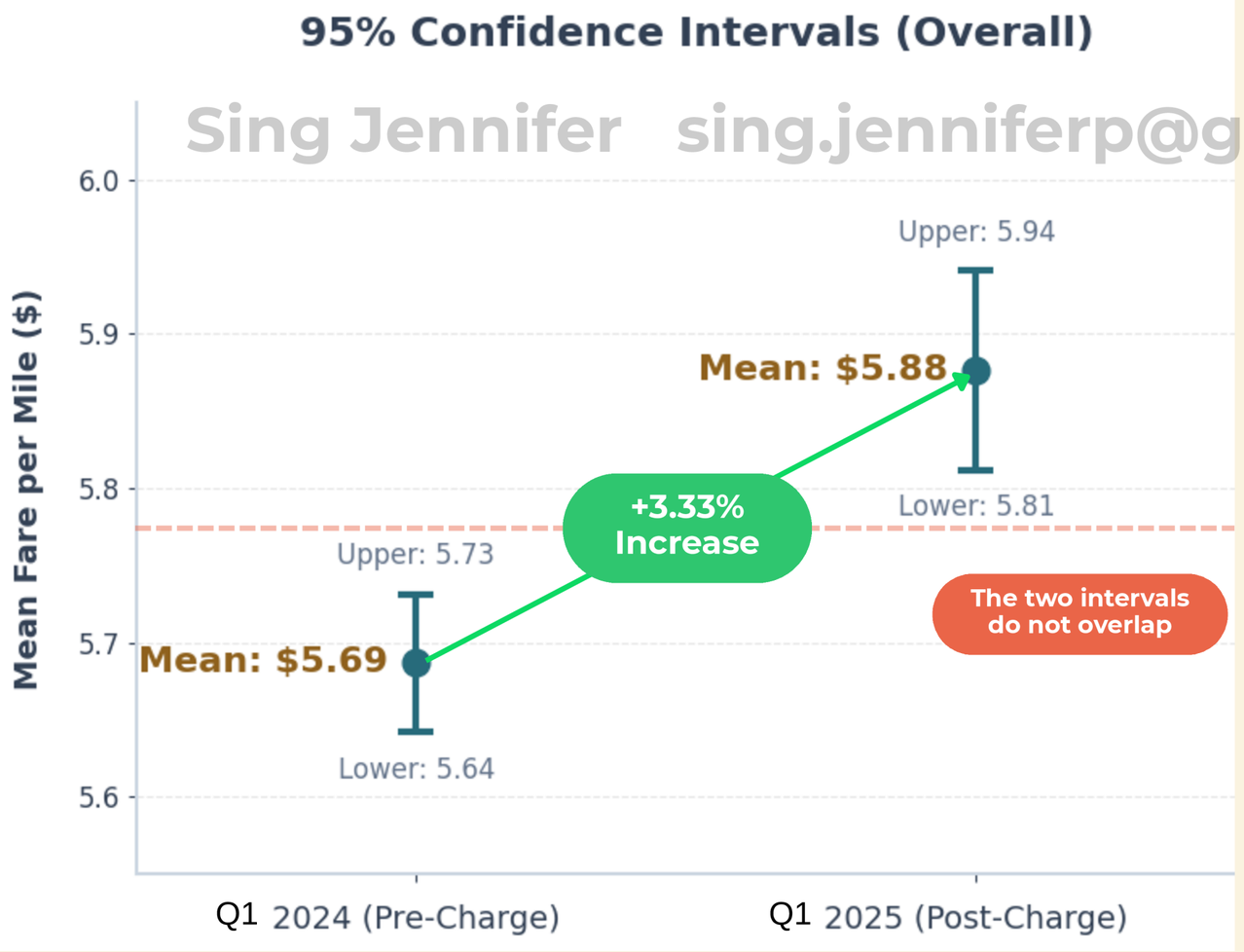
Service Efficiency: Fare-Per-Mile Growth (2024 vs. 2025) Confidence Interval

A +3.33% increase in mean fare per mile, driven by significant growth in FHVHV



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This graph was generated using Python with the assistance of GenAI

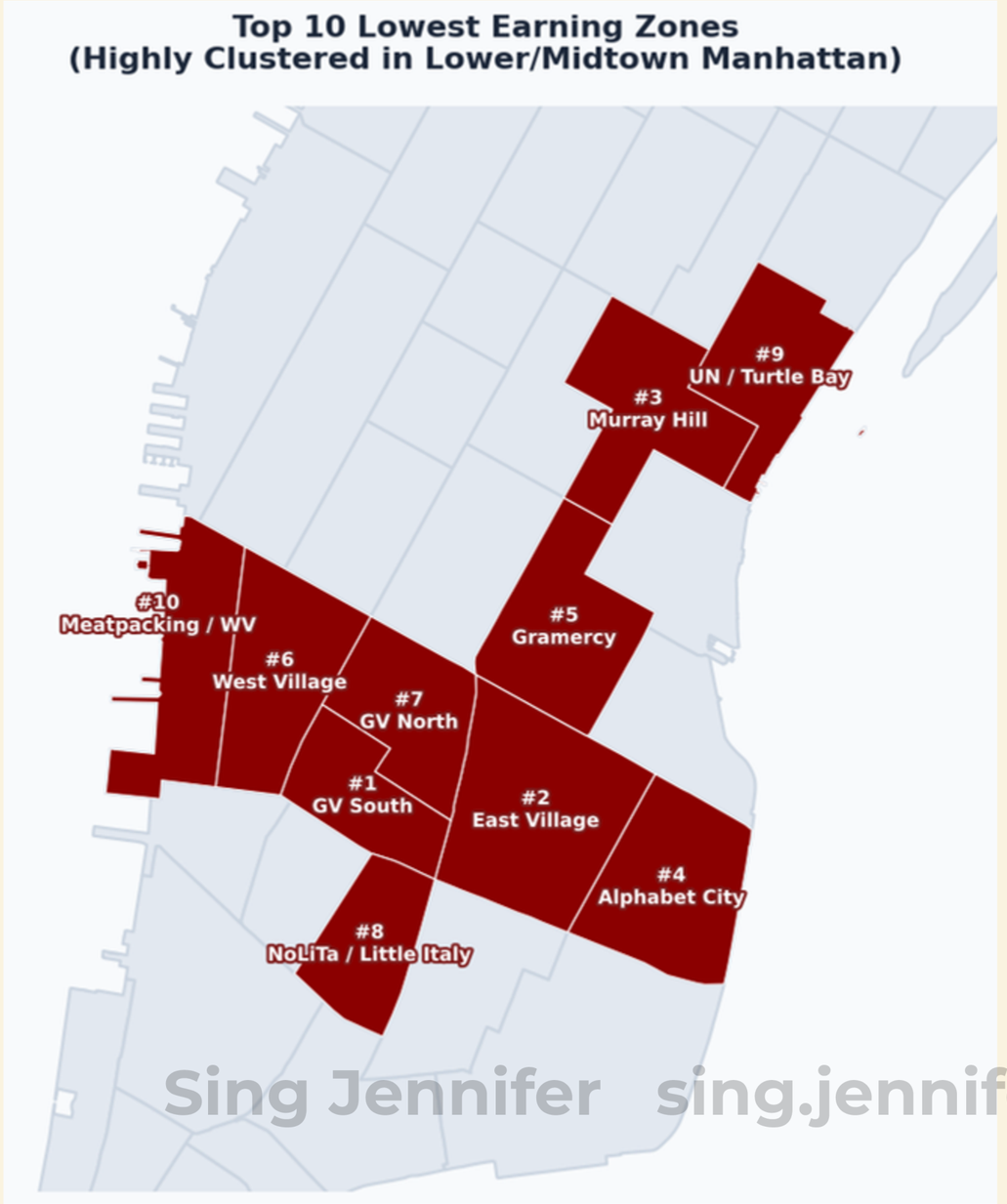
Service	2024 Mean	2025 Mean	Change	Statistical Significance (p)
	$7.27	$7.15	-1.66%	Not Significant
	$7.11	$7.64	+7.47%	Not Significant
	$6.53	**$7.16**	+9.69%	**Significant** ($p < 0.0001$)

Despite Green taxi's +7.47% growth, only FHVHV's growth is statistically significant.

Note: Mean Fare per Mile = (Total Fare / Total Miles) per day.
This metric is consistent across both the CI and T-Test.

Geographic Disparities: Driver Pay by CRZ Zone (One-way ANOVA)

Driver pay is not equal across zones. Tukey HSD analysis identifies the 10 lowest-earning areas.



#	Zone ID	Zone Name
1	114	Greenwich Village South
2	79	East Village
3	170	Murray Hill
4	4	Alphabet City
5	107	Gramercy
6	249	West Village
7	113	Greenwich Village North
8	144	Little Italy/NoLiTa
9	233	UN/Turtle Bay South
10	158	Meatpacking/West Village West

One-Way ANOVA Results

Pay differs significantly by zone.
(Reject H_0 | $F = 28.46$ | $p < 0.0001$)

Failed Levene's Test

ANOVA and Tukey HSD remain robust due to large, balanced sample sizes across zones (n = 1126 to 1379).

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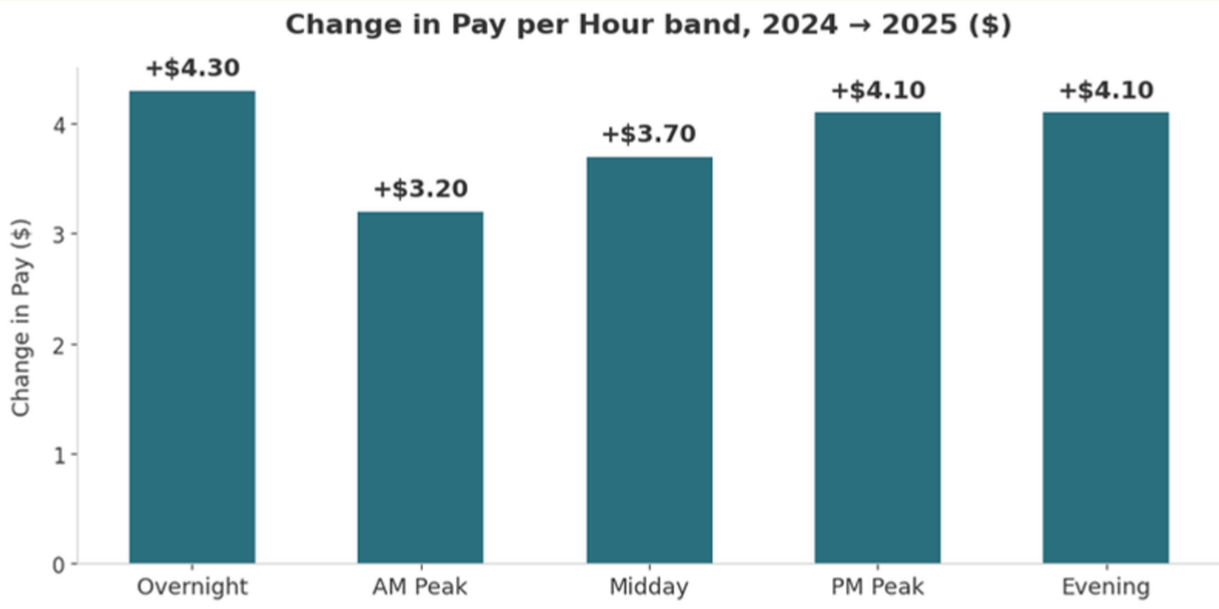
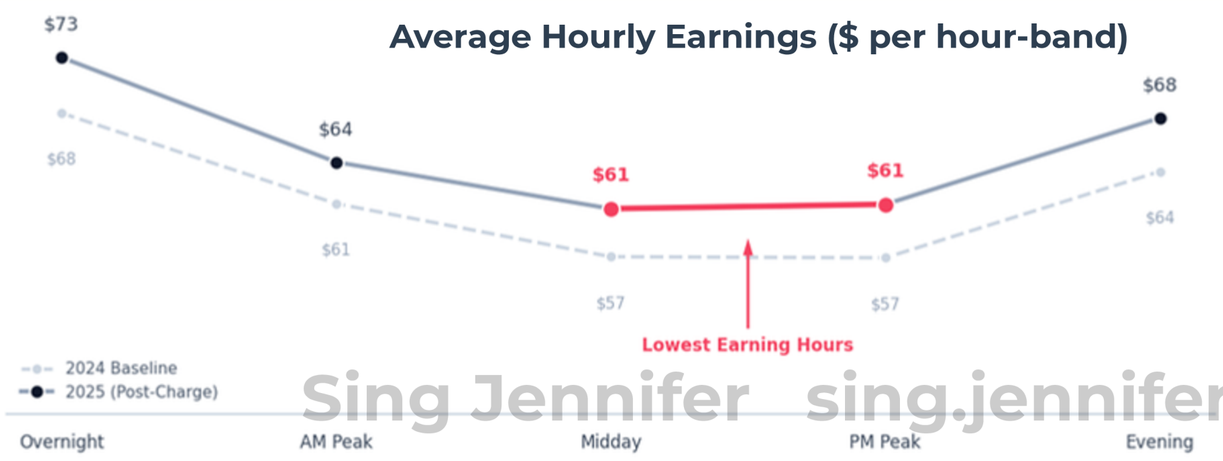
Earning Resilience: Time-of-Day Impact (Two-Way ANOVA)

Driver pay increased uniformly across all hours; time-based incentives should target Midday and PM Peaks.





Uniform Pay Increase

Driver pay on CRZ trips rose by **+$3.20 to +$4.30 per trip-hour (p < 0.001)**, **evenly across all hour bands (p = 0.85).**

Lowest Earning Hour band

Despite the raise, drivers still earn $12/ hour less in the middle of the day compared to overnight

Hour bands difference Highly significant. (p < 0.05) Earnings vary a lot by time of day

Target Incentives

Deploy future retention incentives prioritise towards the Midday & PM Peak windows.

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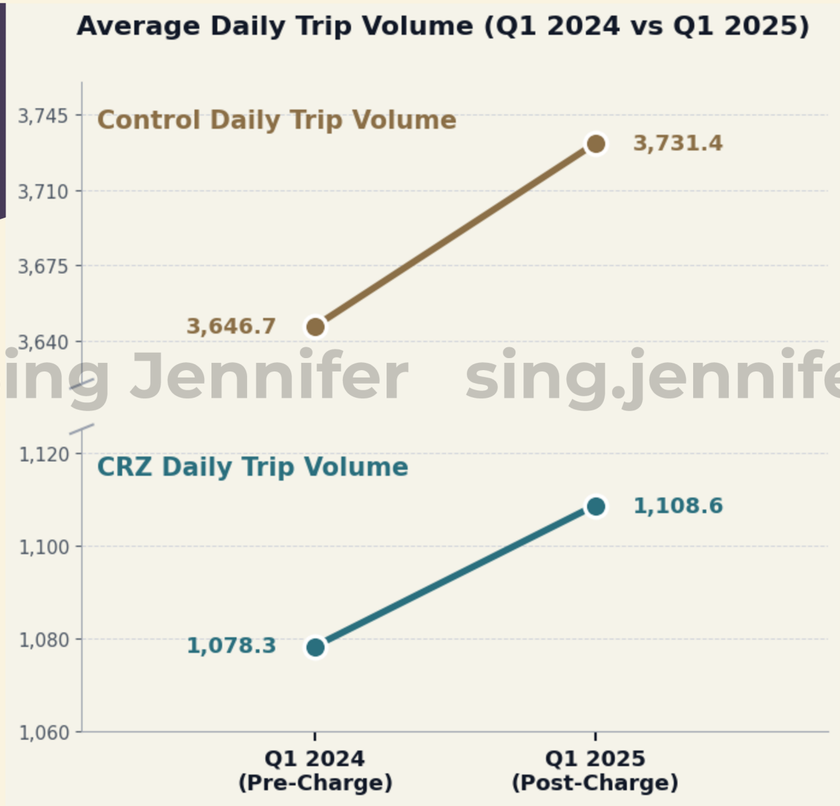
Economic Impact: Trip Volume Assessment

CBD Congestion Fee Shows No Negative Impact on Daily Trip Volume and Proportion of Short-Trips

Trip Volume



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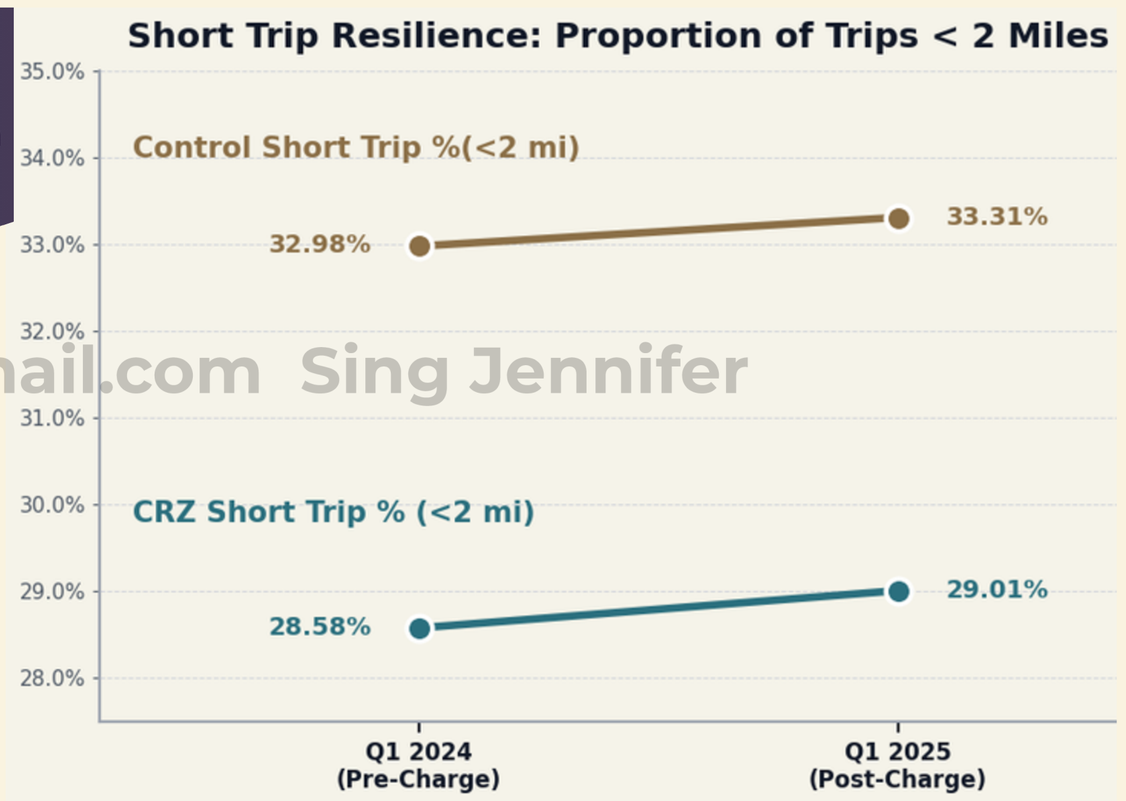
Short Trip Proportion



Verdict: Stable. No Significant Decrease

Passengers continue booking vehicle trips through the congestion zone despite the added fee.
Maintain Manhattan dispatch and peak-hour staging; trip volume remains robust.

Test: T-stat: 1.04 p: 0.85 *Control T-stat: 1.31 p: 0.90*

Verdict: Stable. No Significant Decrease

Short-trip demand inside the CRZ remains resilient. Passengers continue booking <2-mile rides despite the flat fee. Maintain high-density staging; fee shows limited negative effect.

Test: Z-stat: 2.09 p: 0.98 *Control Z-stat: 2.83 p: 0.99*

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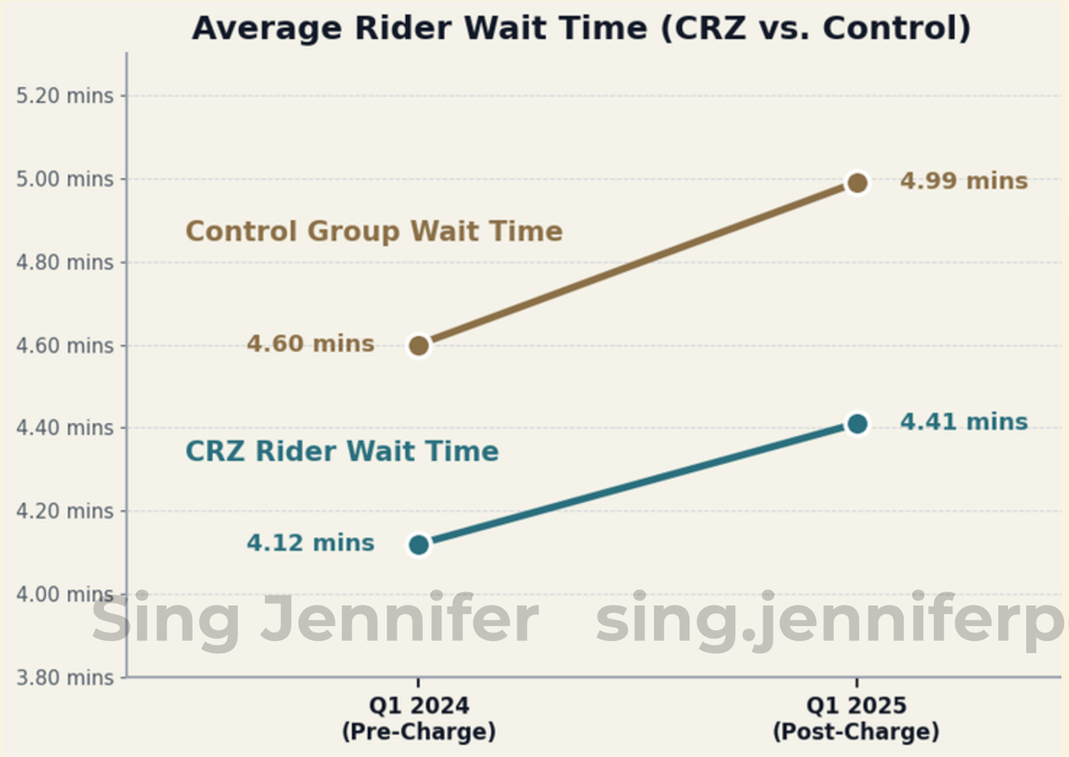
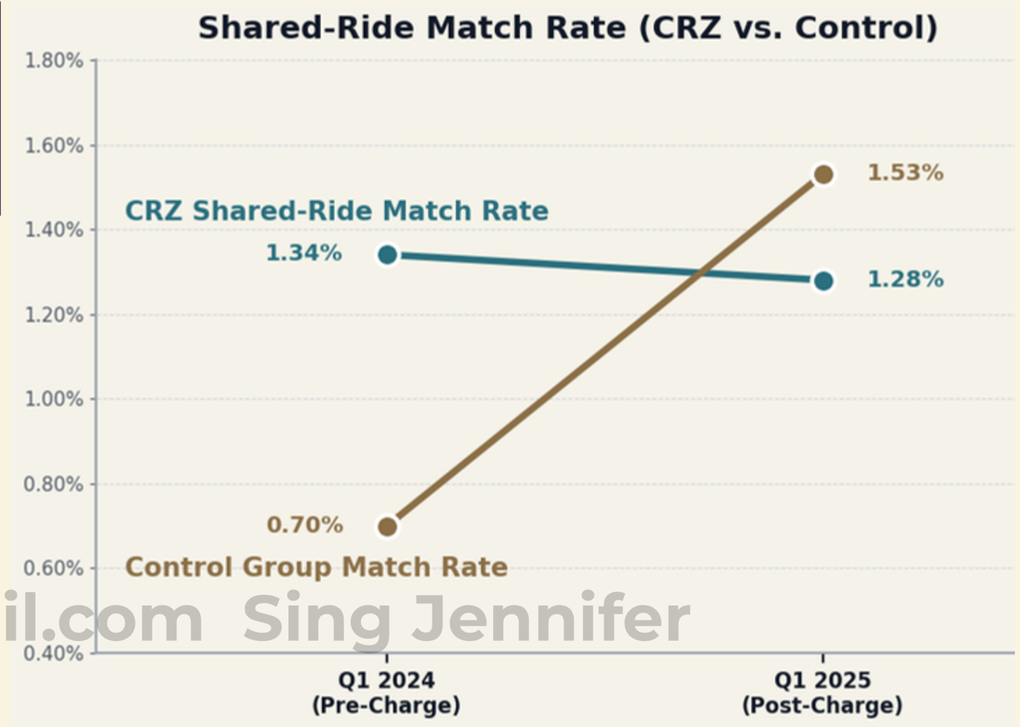
Service Quality: Wait-time & Pooling Assesment

CBD Congestion Fee shows no adverse impact on passenger SLA and proportion of shared-rides

Wait Time



Shared-Ride Proportion



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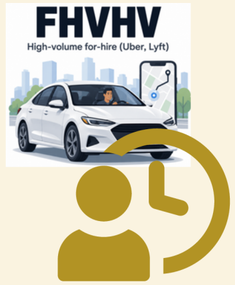

Verdict: Elevated Waits (Network-Wide Issue)

Average rider wait times rose significantly both inside and outside the CRZ. This indicates a broader supply contraction across NYC, not an isolated effect of the congestion fee.

Test: z-stat: 16.77 p < 0.05 *Control z-stat: 40.22 p < 0.05*

Verdict: Stable. Pooling Resillient

Shared-ride match rates inside the CRZ show no significant decline. Passengers continue pooling trips, sustaining operational efficiency and match liquidity despite the congestion fee.

Test: z-stat: -0..94 p = 0.17 *Control z-stat: 28.46 p=1.00*

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Strategic Zone Interventions: K-Means Fleet Clustering

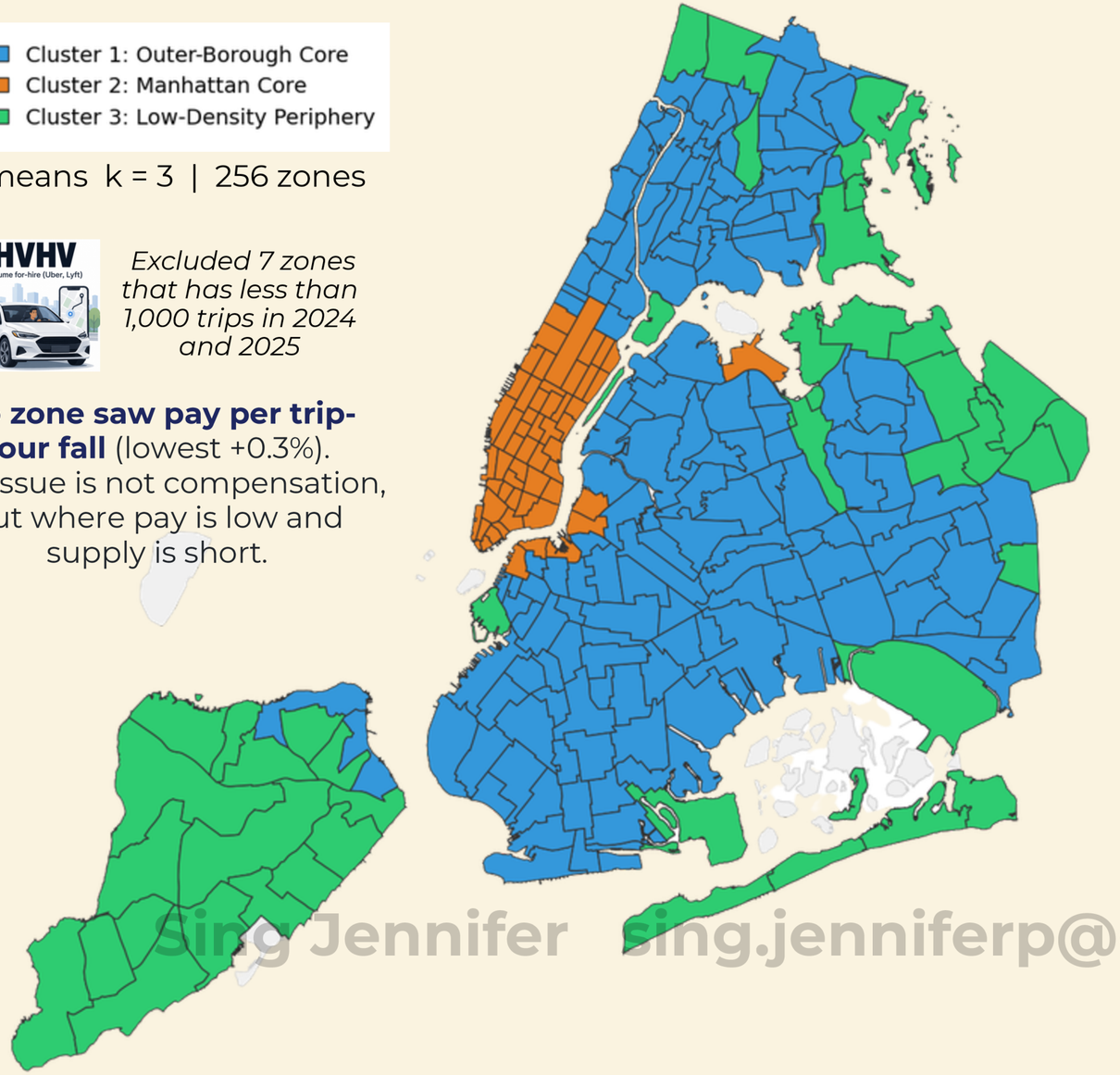
Mapping 256 active NYC zones to isolate low-pay and supply-short margins

Cluster 1: Outer-Borough Core
Cluster 2: Manhattan Core
Cluster 3: Low-Density Periphery

K-means k = 3 | 256 zones

Excluded 7 zones that has less than 1,000 trips in 2024 and 2025

No zone saw pay per trip-hour fall (lowest +0.3%). The issue is not compensation, but where pay is low and supply is short.



C1 — 154 Zones

Pay / trip-hr	$60.1	Lowest
Pay growth	+2.9%	
Avg wait	283 s	
Trip growth	+5.2%	

All 38 Charged Zones are in C2

C2 — 55 Zones

Pay / trip-hr	$64.6	
Pay growth	+5.9%	Highest
Avg wait	262 s	Shortest
Trip growth	−3.4%	Decline

C3 — 47 Zones

Pay / trip-hr	$69.4	Long Trips
Pay growth	+3.1%	
Avg wait	372 s	Longest
Trip growth	+7.8%	

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Intervention Strategies for Cluster C1 & C3

Addressing low pay and long waits in Outer Core & Periphery

C1

Lowest Pay per Trip Hour

Strategy: Pay-side incentives (retention bonuses or targeted subsidies) to lift earnings resilience

Zone (Brooklyn)	$ / trip-hr
Borough Park	54.46
East Flatbush/Farragut	55.13
Prospect-Lefferts Gardens	55.23
Erasmus	55.34
East Flatbush/Remsen Village ★	55.38
South Williamsburg	55.44
Brownsville ★	55.82

C3

Longest Wait Time

Strategy: Increase driver supply (fleet redistribution, targeted bonuses for pickups in periphery

Zone	Excess wait time total /day
Far Rockaway	3,628 mins
Hammels/Arverne	2,896 mins
Flushing Meadows-Corona Park	1,895 mins
Bloomfield/Emerson Hill	1,545 mins
Heartland Village/Todt Hill	1,133 mins
Rockaway Park	1,086 mins
Grymes Hill/Clifton	969 mins

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★ Low pay + long waits: need support on income and supply.

C2 (Manhattan core): no action needed — highest pay growth, shortest waits; monitor trip decline
Airports Excluded: Waits due to pickup queues, not driver shortage.
Pay per Trip-Hour Metric: Counts only passenger time; real hourly earnings are lower.

Fleet Operator Recommendations & Regulatory Guidance

Actionable recommendations for operators, with guidance for regulators

FLEET OPERATOR RECOMMENDATIONS

METRIC	WHAT WE FOUND	RECOMMENDATION
Driver earnings	Pay rose: +$0.48/trip	**No CRZ-wide subsidy; temporary zone-specific incentives only where needed**
Trip volume	No significant drop: charged zone 1,078 → 1,109 trips/day (p = 0.85)	**No intervention now; keep monitoring volume by zone**
Short-trip share	Unchanged at ~29% (p = 0.98); short trips + $0.28 vs long +$0.48 per trip	**Keep the flat fee; keep tracking short-trip share**
Service quality	Wait rose everywhere: charged 4.12 → 4.41, control 4.60 → 4.99 min	**Network-wide issue: regulator monitors; operators reposition supply**
Marginal zones	10 Brooklyn zones pay least ($54-56/hr); Brownsville and E. Flatbush/Remsen Village also have long waits	**Target incentives and repositioning at these two first**

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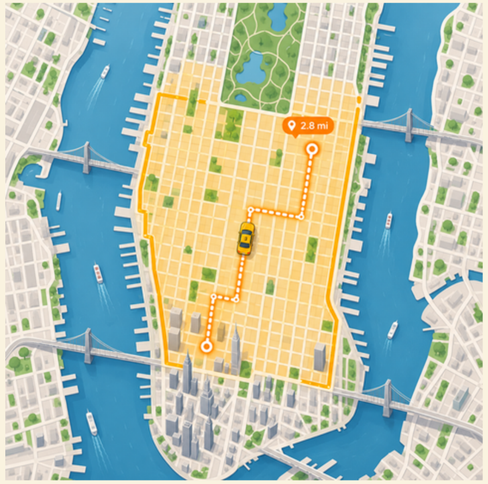
REGULATORY GUIDANCE

1 **Keep the flat $1.50 fee**
No earnings or behaviour problem big enough to justify a redesign yet.

2 **Monitor fee as % of fare**
Track by distance band. If short trips are clearly over-burdened, pilot a hybrid fee first.

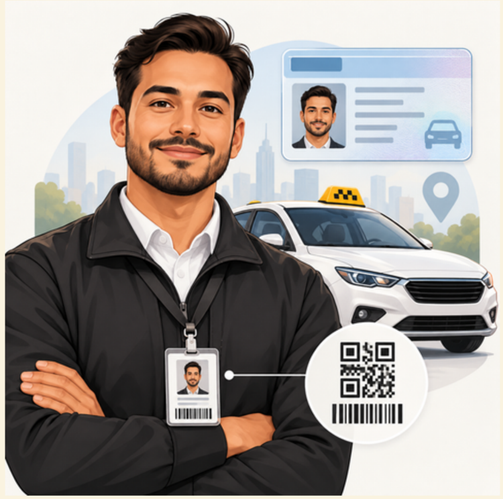
3 **Don't adopt a per-mile fee**
Total distance + congestion caused; distance inside the CRZ isn't measured yet.

Limitations and How to Improve it



Distance travelled inside the zone

Request-level records, including declined offers, abandoned requests, cancellations



Driver identifier

Future researches can improve these aspects to come out with better results

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Idle and cruising time

Distance and fare for FHV

Driver pay for yellow, green and FHV

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Thank you

Use of AI Tools

This slides incorporated the use of ChatGPT and CanvaAI for generating illustrative images and Google Gemini for summarising ideas, finding options for slide titles, checking cohesion across slides, and spotting incorrect grammar.

All AI-generated outputs were critically reviewed, adapted, o ensure accuracy, relevance, and integrity.